Cycle Test 2: worked solutions

14 Oct 2025 · 20 marks · 1 hour · CEPC22

Q1

3 marks

> A horizontal venturimeter with inlet diameter 20 cm and throat diameter 10 cm is used to measure the flow of oil of specific gravity 0.8. The discharge of oil is 60 litres/s. Find the reading of the oil–mercury differential manometer in cm. Take $C_d = 0.98$.

$a_1 = \pi/4 \times 0.2^2 = 0.031416\ m^2$; $a_2 = \pi/4 \times 0.1^2 = 0.007854\ m^2$

$Q = C_d\, a_1 a_2\, \sqrt{(2gh)} / \sqrt{(a_1^2 - a_2^2)}$

$0.06 = 0.98 \times (0.031416 \times 0.007854) \times \sqrt{(2 \times 9.81 \times h)} / \sqrt{(0.031416^2 - 0.007854^2)}$

$0.06 = 0.98 \times 0.00024674 \times \sqrt{(19.62\ h)} / 0.030418 = 0.0079493\ \sqrt{(19.62\ h)}$

$\sqrt{(19.62\ h)} = 7.548 \rightarrow h =$ **2.904 m of oil**

For an oil–mercury differential manometer with reading x:

$h = x\,(S_{Hg}/S_{oil} - 1) = x\,(13.6/0.8 - 1) = 16\ x$

$x = 2.904 / 16 = 0.1815\ m =$ **18.15 cm of mercury**

Q2

2 marks

> Define kinetic energy correction factor, coefficient of discharge, coefficient of contraction and coefficient of velocity.

- **Kinetic energy correction factor (α)**: ratio of the **actual kinetic energy** of the flow per second (from the real velocity distribution) to the kinetic energy calculated using the **average velocity**: $\alpha = (1/AV^3) \int u^3\, dA$. α = 2 for laminar pipe flow, about 1.03–1.06 for turbulent flow.

- **Coefficient of discharge (C_d)**: actual discharge ÷ theoretical discharge. $C_d = C_c \times C_v$ (about 0.6–0.64 for a sharp orifice).
- **Coefficient of contraction (C_c)**: area of the jet at the **vena contracta** ÷ area of the orifice (about 0.61–0.69).
- **Coefficient of velocity (C_v)**: actual velocity of the jet at the vena contracta ÷ theoretical velocity $\sqrt{(2gH)}$ (about 0.95–0.99).

Q3

2 marks

> Derive the relationship between shear stress and pressure gradient in laminar flow.

Consider a small rectangular fluid element of length Δx, height Δy and unit width in steady, uniform laminar flow in the x direction. Forces on it in x:

- pressure on the left face: $p\,\Delta y$; on the right face: $-(p + \partial p/\partial x\,\Delta x)\,\Delta y$
- shear on the bottom face: $-\tau\,\Delta x$; on the top face: $+(\tau + \partial \tau/\partial y\,\Delta y)\,\Delta x$

Steady uniform flow → no acceleration, so the net force is zero:

$$p\,\Delta y - (p + \partial p/\partial x\,\Delta x)\,\Delta y - \tau\,\Delta x + (\tau + \partial \tau/\partial y\,\Delta y)\,\Delta x = 0$$

$$-\partial p/\partial x\,\Delta x\,\Delta y + \partial \tau/\partial y\,\Delta x\,\Delta y = 0 \Longrightarrow \boxed{\partial p/\partial x = \partial \tau/\partial y}$$

The **pressure gradient in the flow direction equals the shear-stress gradient across the flow**.

Q4

2 marks

> Derive the velocity distribution in laminar flow between parallel plates when both plates are at rest.

Let the plates be a distance t apart, y measured from the lower plate. From Q3, $\partial p/\partial x = \partial \tau/\partial y$, and for laminar flow $\tau = \mu\, du/dy$:

$$\partial p/\partial x = \mu\, d^2u/dy^2$$

Since $\partial p/\partial x$ is constant across the section, integrate twice:

$$du/dy = (1/\mu)(\partial p/\partial x)\, y + C_1;\ u = (1/2\mu)(\partial p/\partial x)\, y^2 + C_1 y + C_2$$

Boundary conditions (no slip): $u = 0$ at $y = 0 \rightarrow C_2 = 0$; $u = 0$ at $y = t \rightarrow C_1 = -(1/2\mu)(\partial p/\partial x)\, t$.

$$\boxed{u = (1/2\mu)(-\partial p/\partial x)(t\, y - y^2)}$$

The profile is a **parabola**. Maximum at $y = t/2$: $u_{max} = (1/8\mu)(-\partial p/\partial x)\, t^2$. Average velocity $\bar{u} = (1/12\mu)(-\partial p/\partial x)\, t^2$, so $\mathbf{u_{max} = 1.5\ \bar{u}}$. Pressure drop over length L: $\Delta p = 12\, \mu\, \bar{u}\, L / t^2$.

Q5

3 marks

> Two parallel plates 0.1 m apart have laminar flow of oil between them with a maximum velocity of 1.5 m/s. Calculate the discharge per metre width, the shear stress at the plates, the difference in pressure between two points 20 m apart, the velocity gradient at the plates and the velocity 0.02 m from the plate. $\mu = 2.453$ N·s/m².

$t = 0.1$ m, $u_{max} = 1.5$ m/s, $\mu = 2.453$ N·s/m².

Mean velocity $\bar{u} = u_{max}/1.5 = 1.0$ m/s → **discharge $q = \bar{u} \times t = 1.0 \times 0.1 = 0.1$ m³/s per m width**

Pressure gradient: $u_{max} = (1/8\mu)(-\partial p/\partial x)t^2 \rightarrow -\partial p/\partial x = 8 \times 2.453 \times 1.5 / 0.1^2 = 2943.6$ N/m³

Shear stress at the plates $\tau_0 = (-\partial p/\partial x)(t/2) = 2943.6 \times 0.05 =$ **147.18 N/m²**

Pressure difference over 20 m $\Delta p = 2943.6 \times 20 =$ **58,872 Pa ≈ 58.9 kPa**

Velocity gradient at the plates $du/dy = \tau_0/\mu = 147.18 / 2.453 =$ **60 s⁻¹**

Velocity at y = 0.02 m $u = (1/2\mu)(-\partial p/\partial x)(ty - y^2) = (2943.6 / 4.906)(0.1 \times 0.02 - 0.02^2) = 600 \times 0.0016 =$ **0.96 m/s**

Q6

3 marks

> Derive the Darcy–Weisbach equation for flow through a circular pipe.

Consider steady flow through a horizontal pipe of diameter d between sections 1 and 2, a distance L apart, with mean velocity V and pressures p_1, p_2.

Bernoulli (same velocity and level): $h_f = (p_1 - p_2)/\rho g$.

Force balance on the fluid between the sections: the net pressure force equals the frictional resistance at the wall. Following Froude's experiments, frictional resistance per unit wetted area $\propto V^2$, i.e. $= f'V^2$, where f' is the frictional resistance per unit area at unit velocity.

$$(p_1 - p_2) \times \pi d^2/4 = f' \times (\pi d \times L) \times V^2$$

$$(p_1 - p_2)/\rho g = (4\, f'\, L\, V^2)/(\rho\, g\, d)$$

Putting $f'/\rho = f/2$ (f = coefficient of friction):

$$\boxed{h_f = 4\, f\, L\, V^2 / (2\, g\, d)}$$

Writing the friction factor $f_D = 4f$ (Darcy friction factor, $f_D = 64/Re$ for laminar flow): $\mathbf{h_f = f_D\, L\, V^2 / (2\, g\, d)}$.

Q7

2 marks

> At a sudden enlargement of a water main from 240 mm to 480 mm diameter, the hydraulic gradient rises by 10 mm. Estimate the rate of flow.

Area ratio = $(480/240)^2 = 4$, so $\mathbf{V_1 = 4V_2}$. Bernoulli with the enlargement loss $h_L = (V_1 - V_2)^2/2g$:

$p_1/\rho g + V_1^2/2g = p_2/\rho g + V_2^2/2g + (V_1 - V_2)^2/2g$

Rise in hydraulic gradient = $(p_2 - p_1)/\rho g = [V_1^2 - V_2^2 - (V_1 - V_2)^2]/2g = 2V_2(V_1 - V_2)/2g$

With $V_1 = 4V_2$: $0.010 = 2V_2(3V_2)/(2 \times 9.81) = 6V_2^2/19.62$

$V_2 = \sqrt{(0.0327)} = 0.1808$ m/s ($V_1 = 0.723$ m/s)

$Q = \pi/4 \times 0.48^2 \times 0.1808 =$ **$0.0327\ m^3/s \approx 32.7$ litres/s**

Q8

3 marks

> A pipeline of 0.6 m diameter is 1.5 km long. To increase the discharge, another line of the same diameter is introduced parallel to the first in the second half of the length. Neglecting minor losses, find the increase in discharge if f = 0.01. The head at inlet is 300 mm.

Using $h_f = 4fLV^2/(2gd)$, with f = 0.01, d = 0.6 m, L = 1500 m, H = 0.3 m. Area $A = \pi/4 \times 0.6^2 = 0.2827\ m^2$.

Before (single pipe):

$0.3 = 4 \times 0.01 \times 1500 \times V_1^2 / (2 \times 9.81 \times 0.6) = 5.097\ V_1^2$

$V_1 = 0.2426$ m/s → $Q_1 = 0.2827 \times 0.2426 =$ **$0.0686\ m^3/s$**

After (parallel pipe in the second 750 m): the full flow Q_2 (velocity V) runs in the first half; each parallel pipe carries $Q_2/2$ (velocity V/2) in the second half.

$H = 4f(750)V^2/(2gd) + 4f(750)(V/2)^2/(2gd) = (4f \times 750 / 2gd) \times 1.25\ V^2$

$0.3 = (4 \times 0.01 \times 750 / 11.772) \times 1.25\ V^2 = 3.186\ V^2$

$V = 0.3069$ m/s → $Q_2 = 0.2827 \times 0.3069 =$ **$0.0868\ m^3/s$**

Increase in discharge = 0.0868 − 0.0686 =

$0.0182\ m^3/s$ (≈ 18.2 l/s, a 26.5% increase)

(In general $Q_2/Q_1 = \sqrt{(8/5)} = 1.265$. If the inlet head is meant to be 30 m, as in the textbook version of this problem, all discharges are 10 times larger: $Q_1 = 0.686$, $Q_2 = 0.868$, increase $0.182\ m^3/s$.)